What surprised me most about the cloud deployment process was how many setup steps were required compared to running the application locally. With Docker Compose, I could start the frontend and backend together using one command. Deploying to Google Cloud required me to enable APIs, connect my GitHub repository, create separate Cloud Build configuration files, configure deployment triggers, and deploy both services individually. Once everything was configured correctly, however, Cloud Build made the process much easier by automatically building and deploying the application.

Cloud Run's serverless model is different from running containers locally because Google manages the infrastructure behind the application. Docker Compose runs the containers directly on my computer, meaning my computer must remain active for the application to work. Cloud Run hosts the containers online, starts instances when requests are received, and can scale them down when the application is not being used.

The main advantages of cloud deployment are accessibility, scalability, and consistency. Users can access the application from anywhere through a public URL, while Cloud Run can automatically adjust to changes in traffic. Containerization also helps ensure that the application runs consistently across different environments.